
A D V A N C E D D A T A A N A L Y S I S I N R

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D A N I E L L E E L D O R , S H I R B E R N S T E I N A N D
A M I T H A R I T A N

L I N K T O C O D E : G I T H U B 

S P R I N G S E M E S T E E R , 2 0 2 2 - 2 0 2 3

1. Introduction

The general problem area that our analysis contributes to is assumption making according to past events, specifically with recruitment of at-risk youth. Serving in the army is important, it is not only mandatory, it is part of the Israeli culture and existence. The service is a teachable experience that our whole society goes through and is dependent on, many use this time in life to grow as people and encounter new roots and obstacles which may be very helpful later in life. According to Dar & Kimhi's article, serving in the IDF helps strengthen feelings of independence and self control, and improves awareness of themselves and their surroundings (Dar & Kimhi, 2000). Answering our questions is hard because finding the specific features leading to people's future decisions and actions may be a difficult task in the area of social studies due to the fact that people have many different pasts and many features in life, and choosing only multiple to put a finger on their reactions in the future can be very complicated. Furthermore, not only that our data contains approximately only 200 teens that answered our question in the questionnaire, only one sixth did not recruit to the army, making this data imbalanced. Our data science questions are:

Exploratory: What are the associations between army service and traumas, goals, life skills, and education among at-risk youth?

We will use logistic regressions, to find the correlation and connection between the different relevant feature families and army service.

Predictive: Can we predict the likelihood of at-risk youth engaging in army service based on their reported traumas, goals, life skills, and education?

We will try to do some feature engineering, mark the most important features and fix the imbalanced data using smart multiplication functions. Then, we will try to predict with different models.

Hypothesis: According to previous articles, optimism and readiness for an independent life in the future may help in indicating if a specific teen will recruit in the future (Sulimani-Aidan et al., 2013). Based on these prior findings, we are expecting to see a positive correlation between backgrounds that engage with high hopes, expertise and higher levels of education to higher levels of recruitment, and negative correlation between childhood traumas and recruitment.

2. Data Overview

The data we are researching is a series of questionnaires at-risk teens were asked to fill out during the span of ten years, researching the estimates of childhood adversity and trauma (e.g. family loss, parental issues, maltreatment, and social difficulties) with long-lasting effects. The data was collected over approximately 10 years in 3 different time periods:

T1 - The first time period, at 14-16 years old

T2 - The second time period, a few years later

T3 - The third and last time period, around 10 years after T1

The feature families in our data that are relevant to the problems we are trying to solve are childhood traumas, hopes and goals, preparations and extracurriculars, skills, and education.

3. Methods and results

Preprocessing: During this process we divided our data to train and test sets with a ratio of 0.8. We preprocessed the data based on our train set: handling NA's values using

different methods, fixing imbalanced data with up-sampling, and tried to do feature engineering. To replace NA values, we explored each column using its current values and other columns of its family in order to understand what value was best filling these missing values. After calculating these values on the train set, we used them to replace the NA's of the test set accordingly. To fix imbalance data, we used random upsampling to fix the group sizes to the largest group in the data frame making class distribution equal. Only one in six teens from our data did not recruit, making this process necessary and critical. We used feature engineering to try and improve our models' truthfulness. We calculated correlation between numerous new feature columns and future recruitment, and found several with fine correlation (~ 0.45) and with a low amount of NA's, however, seeing as finding logic behind them was difficult, we decided to not include these in our models.

Exploratory: We divided our potential features into groups and used logistic regression to explore the relationships between the groups and recruitment through the estimate vector and P-value for significance. We received a variety of estimates, some going along with our hypothesis and others in contradiction to it.

Traumas - only one third of the features came out significant, and these, except for one, showed negative correlation as we expected [Fig. 1].

Hopes and Goals - the feature HopeAll_T1 had a P-value of 0.0541 and an influencing estimate (14.878) as was expected, in contrast to most of the other features in this family who had shown surprising negative correlation with recruitment [Fig. 2].

Skills - half of this feature family was not significant. Moreover, most had quite a low influence on future recruitment, whether the estimate was positive or negative, except for B_life_skills_all who is very significant (0.002) and had shown negative correlation with recruitment in contrast with our expectations.

Education - the education and extracurricular programs had shown poor significance, and as for the estimates, these had low influence on recruitment.

General - Continuing on with our research, we found two more relevant features which were not mentioned in the project proposal: religiousness and gender. These features have a very significant P-value (smaller than $1e-6$) and have positive reliable correlation with recruitment [Fig. 3].

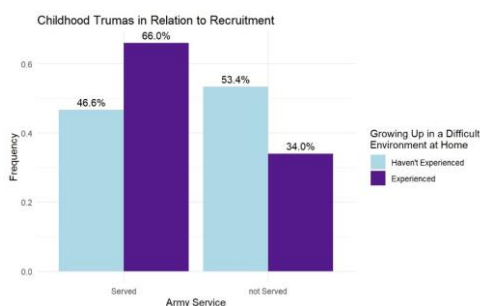


Fig. 1

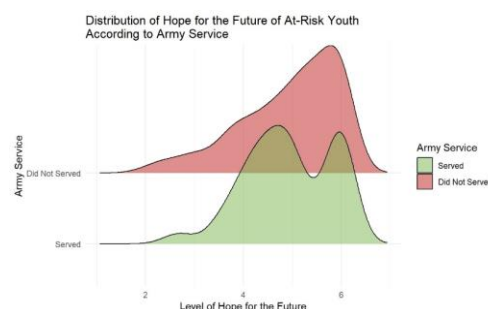


Fig. 2

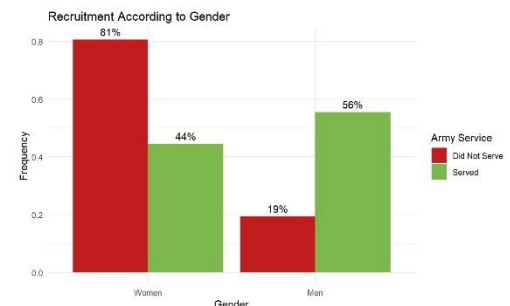
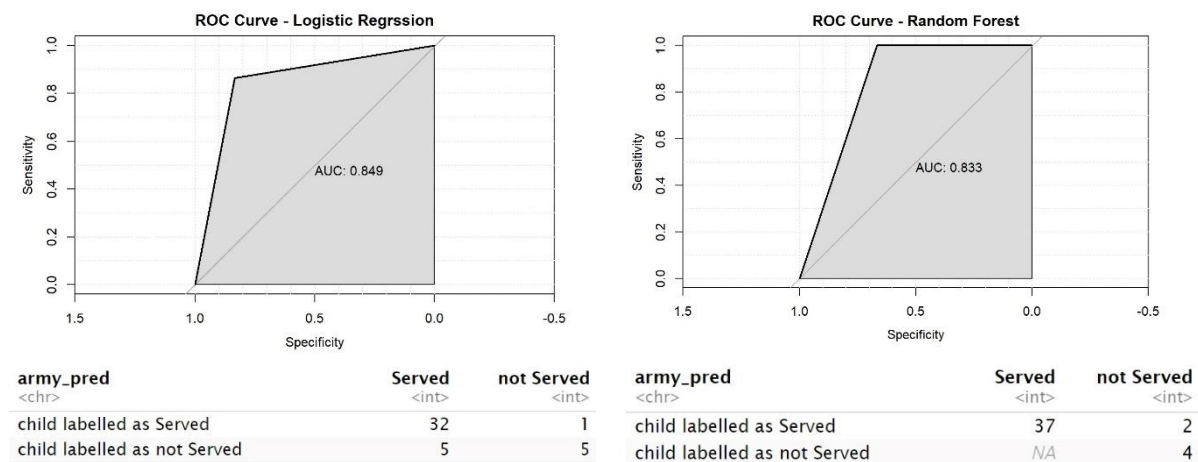


Fig. 3

Predictive: First, we wanted to reduce the number of features we are using in our models to avoid overfitting. We calculated feature importance using SHAP, random forest, etc. To tackle our data from multiple points of view we decided to train two different classification models:

Logistic Regression - for this model we created a recipe that uses multiple relevant columns according to our findings in the exploratory part of our research and with consideration in feature importance. We tried to maximize our AUC-ROC and got a value of 0.849 and an F1 score of 0.914.

Random Forest - in addition to our LR model we decided to train a RF model too to use these aspects of the model: capability of capturing non-linear relationships, insensitivity to outliers, ability to handle complex interactions and dependencies (as relevant for high dimensional datasets), and reducing risk of overfitting. With this model we achieved a AUC-ROC value of 0.833 with an F1 score of 0.974 and a significant P-value of 0.0495.



We have to take into consideration that our scores may be high because the models can easily predict most of the youth who did recruit to the army, and because of the imbalances in our data. In order to balance out our test set we would have had to take rows from our train set, which is already quite small as is, making this process problematic.

From these models we learned that predicting future recruitment of at-risk youth is possible using the data supplied. The features that have shown themselves to be most relevant and contributing to research around this topic are the teen's religion, gender, feelings of preparedness towards the future, some life skills (oddly with a negative correlation), feeling controlless over their life, and having high hopes toward the future. Surprisingly, the features on the other hand that were the least influential were specifically mainly in the trauma feature family and the education and extracurricular feature family.

4. Limitations and Future Work

The data we are working on is difficult to extract results from because of multiple factors: the data was very imbalance (181 teens recruited in contrast to only 32 that did not), many unreasonable columns and with tangled connections to others, many NA's (e.g. the original data contains ~400 teens while only ~200 answered the question on if they were recruited or not), thus creating a situation where there are not enough rows to work with.

If we had an additional month to work on this project we would have wanted to try researching youth in general and comparing and contrasting the features and aspects leading to recruitment with at-risk youth. Moreover, as for the three month range we would have wanted to expand our research to contribute to society among at-risk youth while comparing between different ways of contribution.

Appendix:

Bibliography:

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- Sulimani-Aidan, Y., Benbenishty, R., Dinisman, T., & Zeira, A. (2013). Care leavers in Israel: What contributes to better adjustment to life after care?. *Journal of Social Service Research*, 39(5), 704-718.